Supervised and Unsupervised Approaches to Stellar Classification Using SDSS Photometric Data

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1. Introduction (Diego)

This project uses the SDSS17 Stellar Classification dataset from Kaggle, containing 100,000 observations classified as stars (21,594), galaxies (59,445), or quasars (18,961). Each observation includes 17 features such as photometric measurements, spectroscopic attributes such as redshift, and survey metadata. Our research question is: can celestial objects be accurately classified using photometric features, and do the resulting categories reflect the natural structure in the data? We address this through supervised learning (KNN and SVM), which uses known spectroscopic labels to build predictive models, and unsupervised learning (hierarchical clustering), which examines whether the three object types form distinct groups without label information. PCA is used as a supporting technique for dimensionality reduction and visualization.

2. Methodology

2.1. KNN (Asibong)

K-Nearest Neighbors (KNN) is used as one of the supervised learning methods to classify objects into stars, galaxies, or quasars. This method works by comparing each observation to others in the dataset and assigning a class based on the closest neighbors. The main idea is that observations with similar photometric features, such as magnitudes in different filters, should belong to the same class.

To determine which observations are closest, the Euclidean distance is used:

$$d(x, x_i) = \sqrt{\sum_{j=1}^{p} \left(x_j - x_{i,j}\right)^2}$$

After computing distances, the model selects the k nearest neighbors and assigns a class using majority voting:

$$\hat{y} = \text{mode}\{y_i : i \in N_k(x)\}$$

Where N_k represents the set of the k nearest neighbors.

KNN was chosen because it does not assume any specific shape for the decision boundary, which is useful given the complexity of astronomical data. It can adapt to local patterns in the feature space, especially when relationships between variables are not linear. However, the method is sensitive to irrelevant or unscaled features, which makes preprocessing steps like normalization important. It is also computationally expensive at prediction time since distances must be calculated for every new observation.

2.2. SVM (Diego)

Support Vector Machines seek a globally optimal decision boundary by finding the hyperplane that maximizes the margin between classes. For a binary classification problem, the hyperplane is defined as:

$$f(x) = \beta_0 + \beta_1 x_1 + ... + \beta_p x_p = 0$$

Observations are classified based on which side of the hyperplane they fall on. To handle non-linearly separable data, SVM uses kernel functions that map the original features into a higher-dimensional space where linear separation becomes possible. We selected the Radial Basis Function (RBF) kernel defined as:

$$K(x_i, x_j) = \exp(-\gamma \parallel x_i - x_j \|^2)$$

where γ controls how much influence a single training observation has. The optimization problem minimizes:

$$\min_{\beta, \beta_0} \frac{1}{2} \|\beta\|^2 + C \sum_{i=1}^{n} \xi_i$$

subject to $y_i(\beta_0 + \beta^T \phi(x_i)) \geq 1 - \xi$ and $\xi \geq 0$, where C is the cost parameter controlling the trade-off between a wider margin and fewer misclassifications, and ξ_i are slack variables that allow some observations to fall on the wrong side of the margin. SVM is appropriate for this dataset because the three celestial object classes likely have complex nonlinear boundaries. It's advantage over KNN is that once trained, SVM predictions are computationally efficient and the model generalizes well to unseen data. Its disadvantage is its longer training time.

2.3. PCA (Anthony)

Principal Component Analysis (PCA) finds linear combinations of the original features that capture the maximum variance in the data. The first principal component is defined as:

$$Z_1 = \varphi_{11} X_1 + \varphi_{21} X_2 + ... + \varphi_{p1} X_p$$

where the loadings φ_{j1} are chosen to maximize $\mathrm{Var}(Z_1)$ subject to the constraint $\sum_{j=1}^{p} \varphi_{j1}^2 = 1$. Each subsequent component is orthogonal to the previous ones and captures the next largest direction of variance. PCA is appropriate here as a preprocessing step for clustering because the photometric filters exhibit high multicollinearity, and reducing to a small number of components removes this redundancy while preserving most of the variance. Its advantage is computational efficiency and improved cluster separation in lower-dimensional space. Its disadvantage is that components are linear combinations of features and therefore lose the direct astrophysical interpretability of individual filters.

2.4. Hierarchical Clustering (Asibong)

Hierarchical clustering is used as an unsupervised learning method to explore whether the data naturally forms groups without relying on the class labels. Instead of predicting categories, this method builds a structure that shows how observations cluster together based on similarity.

In this analysis, an agglomerative approach is used. Each observation starts as its own cluster, and clusters are merged step by step. The choice of which clusters to merge depends on a linkage criterion. For example, using complete linkage, the distance between two clusters is defined as:

$$D(A, B) = \max_{x \in A, y \in B} d(x, y)$$

This means that clusters are merged based on the maximum distance between their points. Other linkage methods could produce different clustering structures, but complete linkage tends to create more compact clusters.

Hierarchical clustering was selected because it does not require specifying the number of clusters in advance and provides a dendrogram that shows relationships at different levels of grouping. This is useful for understanding whether stars, galaxies, and quasars form distinct clusters in the feature space. On the downside, the method becomes computationally expensive as the dataset grows, and the final clustering can be sensitive to the choice of linkage method and distance metric.

3. Data Analysis

3.1. Preprocessing (Diego)

The raw dataset contains 100,000 observations and 18 features. We removed nine metadata columns (`obj_ID`, `run_ID`, `rerun_ID`, `cam_col`, `field_ID`, `spec_obj_ID`, `plate`, `MJD`, `fiber_ID`) since these had no use for our project. We also dropped `alpha` and `delta`, which are sky coordinates showing telescope pointing direction rather than object properties. The correlation matrix confirmed near-zero correlation between these positional features and all other variables, and boxplots showed no class separation.

One observation contained outlier values of -9999 in the `u`, `g`, and `z` columns and was thus removed. This left us with 99,999 observations and 6 features: `u`, `g`, `r`, `i`, `z`, and `redshift`. All features were standardized to have a mean of zero and unit variance. This scaling was necessary because magnitude features ranged from 10 to 32, while redshift ranged from 0 to 7. Without scaling, the magnitude features would dominate distance calculations when using clustering methods.

3.2. Exploratory Data Analysis (Diego)

The dataset contains three classes: `GALAXY` (59,445), `QSO` (18,961), and `STAR` (21,594). `GALAXY` accounts for 59% of observations, creating a moderate class imbalance that may bias KNN predictions toward the majority class.

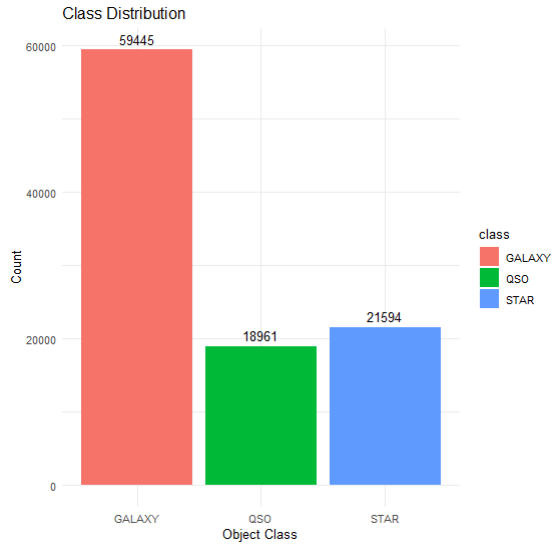

Figure 1: The dataset contains 59,445 galaxies (54.9%), 21,594 stars (21.6%), and 18,961 quasars (19.0%), indicating moderate class imbalance

The correlation matrix revealed high multicollinearity among the five magnitude filters, with pairwise correlations ranging from 0.85 to 0.97. Notably, the strongest correlations occurred between adjacent filters, such as r–i at 0.96, i–z at 0.97, and g–r at 0.93. This multicollinearity motivates the use of PCA for dimensionality reduction prior to clustering.

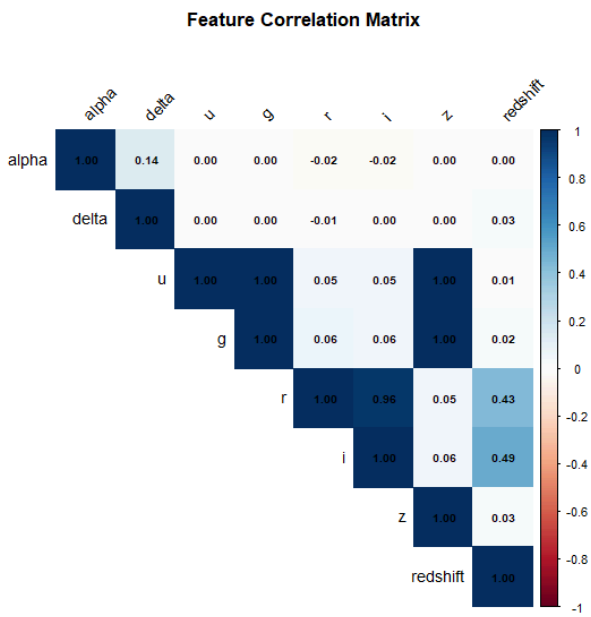

Figure 2: Pairwise correlation among the six retained features. Adjacent magnitude filters show strong multicollinearity, motivating the use of PCA prior to clustering

Redshift emerged as the strongest class separator. Density plots and boxplots showed near complete separation. STAR observations cluster around 0, GALAXY around 0.3 to 0.6, and QSO around 1.5 to 2.2. The magnitude filters r and i showed mild separation: STAR tends toward brighter (lower) magnitudes, whereas u, g, and z exhibited substantial overlap across classes. Although skewness was modest for

all features after sentinel removal, redshift remained the most skewed at 2.52, a result of the long tail from high redshift quasars.

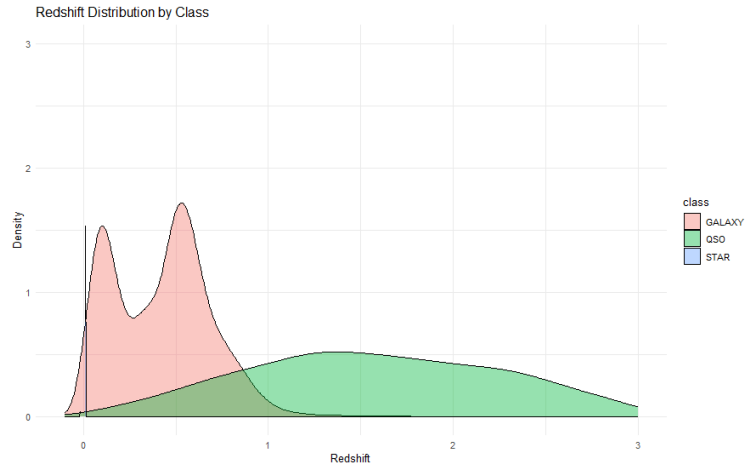

Figure 3: Redshift provides near-complete class separation. Stars cluster near zero, galaxies peak around 0.3-0.6, and quasars spread between 1.0 and 2.5

3.3. Supervised Modeling

3.3.1. KNN (Asibong)

Our goal was to determine how accurately stars, galaxies, and quasars could be classified using supervised learning. Before fitting the KNN model, we removed the class variable from the predictors and separated it as the response variable. Since KNN is a distance-based method, scaling was necessary because the predictors were measured on different scales. We standardized all predictor variables using centering and scaling so that no single variable would dominate the distance calculations.

Using the same random split of 80% training data and 20% testing data, we tested odd values of K from 1 to 21 to find the best tuning parameter. The model achieved its highest testing accuracy at $K = 5$, with an accuracy of 96.06% and a prediction error of 3.94%.

The confusion matrix showed strong classification performance across all three classes. The model correctly classified most galaxies, quasars, and stars, with especially high sensitivity for stars (98.53%) and galaxies (96.31%). Some confusion occurred between galaxies and quasars, which is expected since these objects can share similar photometric characteristics.

```
> # ---- Test set performance ----------------------------------------------------
> # Evaluate model performance using confusion matrix on test data
> cm <- confusionMatrix(knn_pred, test_labels)
> print(cm)
Confusion Matrix and Statistics

          Reference
Prediction GALAXY   QSO  STAR
    GALAXY  11425   275    63
    QSO       155  3551     0
    STAR      283    11  4237

Overall Statistics

               Accuracy : 0.9606
                 95% CI : (0.9579, 0.9633)
    No Information Rate : 0.5932
    P-Value [Acc > NIR] : < 2.2e-16

                  Kappa : 0.9306

 Mcnemar's Test P-Value : < 2.2e-16

Statistics by Class:

                     Class: GALAXY Class: QSO Class: STAR
Sensitivity                 0.9631     0.9255      0.9853
Specificity                 0.9585     0.9904      0.9813
Pos Pred Value              0.9713     0.9582      0.9351
Neg Pred Value              0.9468     0.9824      0.9959
Prevalence                  0.5931     0.1918      0.2150
Detection Rate              0.5713     0.1776      0.2119
Detection Prevalence        0.5881     0.1853      0.2266
Balanced Accuracy           0.9608     0.9579      0.9833
```

Figure 4:

Based on these results, $K = 5$ was selected as the optimal KNN model. Overall, the model achieved very strong predictive performance, suggesting that the SDSS photometric features are highly effective for distinguishing between stars, galaxies, and quasars.

3.3.2. SVM (Diego)

3.3.2.1. Kernel Selection

To select the appropriate kernel, we compared linear, polynomial, and RBF kernels using default parameters on a 15,000-row subsample of the training set. The RBF kernel achieved the highest training accuracy at 96.01%, followed by the linear kernel at 95.59% and the polynomial kernel at 91.70%. Since our objective is to maximize training accuracy in the initial selection, we chose the RBF kernel for all subsequent tuning steps due to its superior performance on this metric

Table 1: Kernel Comparison (Accuracy rounded)

Kernel	Accuracy	Time (sec)
Linear	0.96	2.1
Polynomial	0.92	3.6
Radial	0.96	6.1

3.3.2.2. Hyper-parameter Tuning

We performed a coarse grid search over cost $\in \{0.1, 1, 10, 100\}$ and gamma $\in \{0.01, 0.1, 0.5, 1\}$ using 5-fold cross-validation (CV) on the 15,000 row subsample. The optimal parameters were $C = 100$ and $\gamma = 0.5$, yielding a CV error rate of 2.97%. The lowest errors clustered around high cost and moderate

to high gamma values, indicating the model benefits from a narrow margin and strong penalization of misclassifications.

The full performance metrics for the 16 parameter combinations tested during the coarse grid search, including CV error and dispersion are in **Appendix A**.

3.3.2.3. Test Set Evaluation

Using the optimal parameters, we trained the final SVM on the full training set of $79,999$ observations. The model produced $5,666$ support vectors distributed across three classes: $(2,692)$ for `GALAXY`, $1,988$ for `QSO`, and 986 for `STAR`. On the held-out test set of 20,000 observations, the model achieved a test error rate of 2.83% and an overall accuracy of 97.18%. Per-class accuracy was 97.83% for `GALAXY`, 92.18% for `QSO`, and 99.81% for `STAR`.

Table 2: Confusion Matrix and Classification Performance on Test Set $(N = 20,000)$

Predicted Actual	GALAXY	QSO	STAR
GALAXY	11,606	292	8
QSO	127	3,537	0
STAR	130	8	4,292

3.3.2.4. Full Dataset Refit

As required, we refit the SVM with optimal parameters on the full dataset of $99,999$ observations. The model produced $6,935$ support vector ($3,322$ `GALAXY`, $2,469$ `QSO`, $1,144$ `STAR`), comprising 6.94% of the total data. The relatively low proportion of support vectors indicates the decision boundaries are well-defined, and most observations are not near the class boundaries. `QSO` contributes the most support vectors relative to its class size, confirming that quasars are the most difficult class to separate.

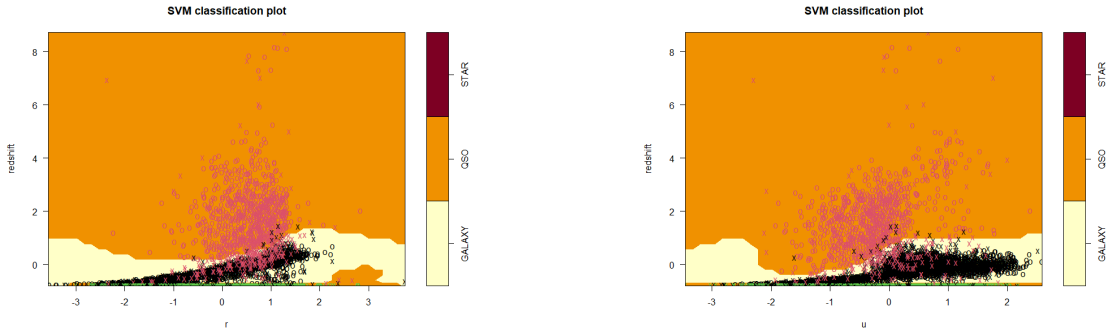

Figure 5: SVM Decision Boundaries: Redshift vs. r-band (left) and u-band (right) magnitudes.

3.4. Unsupervised Modeling

3.4.1. PCA (Anthony)

PCA was applied to the standardized features to reduce dimensionality and identify the dominant directions of variation in the dataset. The first principal component (PC1) accounts for 74.97% of the total variance, while PC2 accounts for 15.62%. Together, the first two components explain 90.59% of the variance, and including PC3 raises this to 97.85%. Given that most of the variance is concentrated

in the first two components, we can reduce the feature space from six dimensions to two with minimal loss of explanatory power.

Table 3: Variance explained by each principal component

Component	Std Dev	Prop. of Variance	Cumulative Prop.
PC1	2.1209	0.7497	0.7497
PC2	0.9682	0.1562	0.9059
PC3	0.6599	0.0726	0.9785
PC4	0.3039	0.0154	0.9939
PC5	0.1504	0.0038	0.9976
PC6	0.1189	0.0024	1.0000

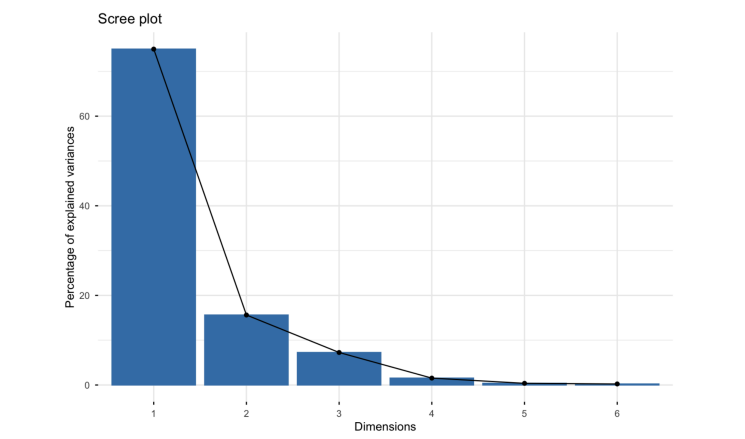

Figure 6: Scree plot of variance explained by each principal component. The elbow after PC2 confirms that two components are sufficient to capture the dominant structure.

Examining the loadings of the first two components reveals which features drive the dominant directions of variation. PC1 loads roughly evenly across all five magnitude filters with similar negative weights, indicating that this component captures overall brightness. PC2 is dominated by redshift (0.8077) and contrasts the bluer filters (u, g) against the redder ones (i, z), reflecting spectral shape and distance.

Table 4: Loadings of the first two principal components

Feature	PC1	PC2
u	−0.3645	−0.4862
g	−0.4418	−0.2643
r	−0.4638	−0.0470
i	−0.4547	0.1009
z	−0.4376	0.1705
redshift	−0.2418	0.8077

Although the original feature space is six-dimensional, the data effectively lives on a two-dimensional manifold. Clustering in this reduced space is therefore both valid and computationally advantageous.

3.4.2. Hierarchical Clustering (Asibong)

Our main goal was to determine whether stars, galaxies, and quasars naturally form groups based on their SDSS observational features. Since hierarchical clustering is distance-based, we first removed the class variable because unsupervised learning should not use true labels during training. We also excluded the response variable and standardized all predictors using z-score scaling because the variables were measured on different numerical scales, and unscaled features could dominate the distance calculations.

Because hierarchical clustering is computationally expensive on large datasets, we randomly sampled 5,000 observations using `set.seed(123)` for reproducibility. To reduce dimensionality while preserving most of the variation, we applied Principal Component Analysis (PCA). The first two principal components explained over 90% of the total variance, so these components were used for clustering.

We tested three linkage methods (complete, average, and single linkage) and evaluated cluster sizes from $K = 2$ to $K = 6$ using silhouette coefficients. The best result came from single linkage with $K = 2$, which produced the highest silhouette score of 0.607. This indicated the strongest cluster separation among all models tested.

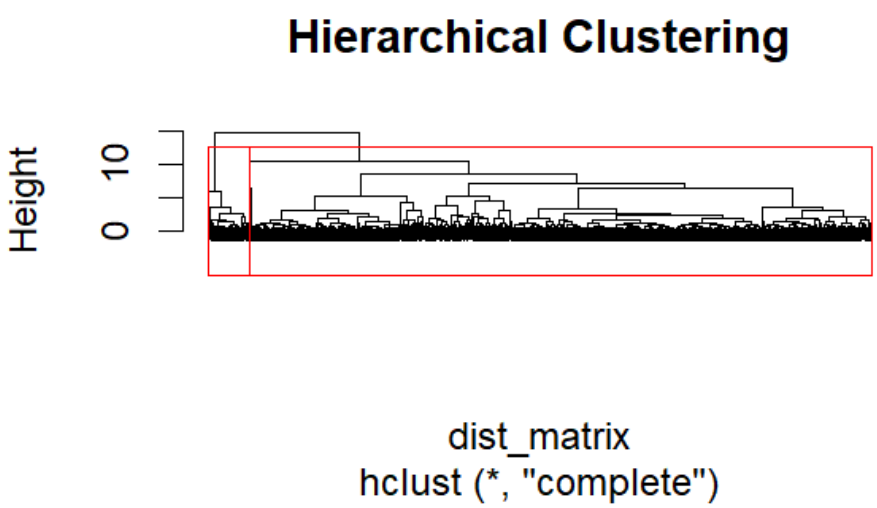

Figure 7: Dendrogram from agglomerative hierarchical clustering with single linkage on a 5,000-observation subsample, projected onto the first two principal components. The optimal cut at $K = 2$ separates stars from galaxies and quasars.

When comparing the clusters to the true class labels, almost all galaxies and quasars were grouped into one cluster, while stars formed the second cluster. This suggests that the photometric features do capture meaningful differences between astronomical object types, although galaxies and quasars appear more similar to each other than to stars.

Based on these results, single linkage hierarchical clustering with $K = 2$ was selected as our best model. Overall, the clustering revealed meaningful structure in the data and showed that stellar objects can be separated reasonably well using observational measurements alone.

4. Conclusion

Our research question asked whether celestial objects could be accurately classified using photometric features, and whether the resulting categories reflect natural structure in the data. The supervised results answer the first half decisively. KNN with $K = 5$ achieved 96.06% test accuracy, and SVM with an RBF kernel achieved 97.18% test accuracy, with SVM outperforming KNN on every class. Both models confirm that photometric features, particularly redshift, provide enough signal to distinguish stars, galaxies, and quasars at high accuracy without spectroscopic input.

The unsupervised results answer the second half with more nuance. Hierarchical clustering with single linkage produced its strongest separation at $K = 2$ rather than $K = 3$, with stars forming one cluster and galaxies and quasars merging into the other. This shows that while photometric features capture meaningful astrophysical structure, the natural separation in the data is between stellar and extragalactic objects rather than between all three labeled classes. Galaxies and quasars share enough photometric similarity that they are not easily distinguishable without supervision.

Several challenges occurred in the analysis. The dataset's size made distance based methods computationally expensive: SVM required subsampling for hyperparameter tuning, and hierarchical clustering required reduction to 5,000 observations. PCA was necessary here, since condensing six features into two components that captured over 90% of the variance and made clustering tractable.

Future work could explore additional clustering methods to better separate galaxies from quasars, or incorporate non-photometric features such as spectroscopic indicators. Overall, the analysis demonstrates that photometric data alone is highly effective for predictive classification, while unsupervised methods recover the broader structure between stellar and extragalactic objects.

Appendix A

Table 5: Coarse Grid Search Results for RBF Kernel Hyperparameters (5-fold CV)

Cost (C)	Gamma (γ)	Mean Error	Dispersion
0.1	0.01	0.1541	0.0047
1.0	0.01	0.0806	0.0013
10.0	0.01	0.0439	0.0018
100.0	0.01	0.0337	0.0025
0.1	0.10	0.0860	0.0029
1.0	0.10	0.0445	0.0013
10.0	0.10	0.0343	0.0025
100.0	0.10	0.0298	0.0024
0.1	0.50	0.0638	0.0017
1.0	0.50	0.0389	0.0015
10.0	0.50	0.0319	0.0024
100.0	0.50	0.0297	0.0026
0.1	1.00	0.0607	0.0032
1.0	1.00	0.0375	0.0012
10.0	1.00	0.0318	0.0021
100.0	1.00	0.0325	0.0010

Appendix B: Supplemental Visualizations

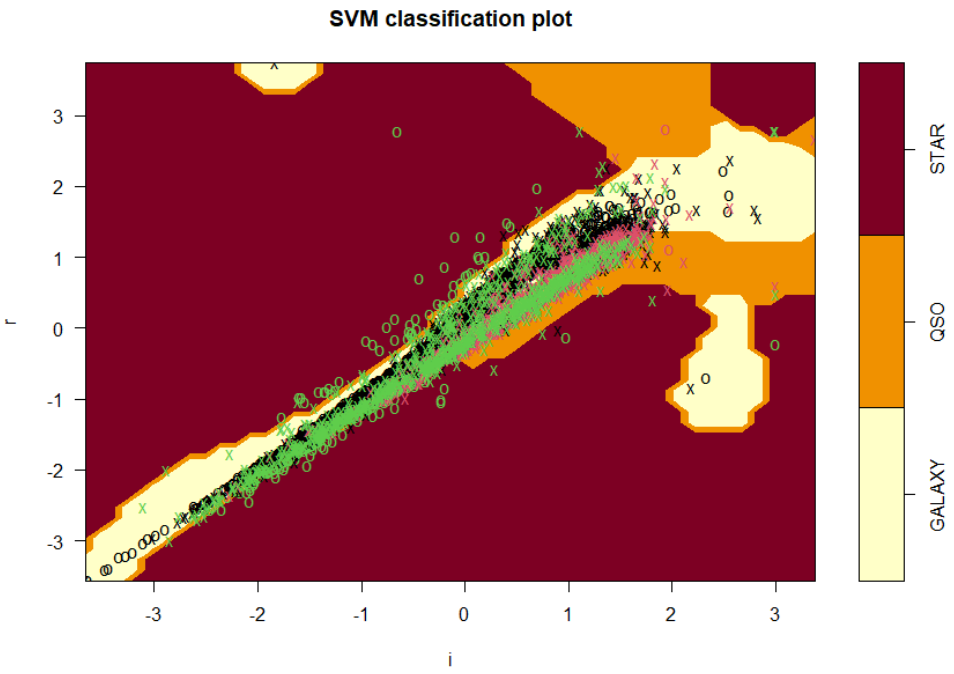

Figure 8: SVM Decision Boundary: r-band vs. i-band magnitudes.

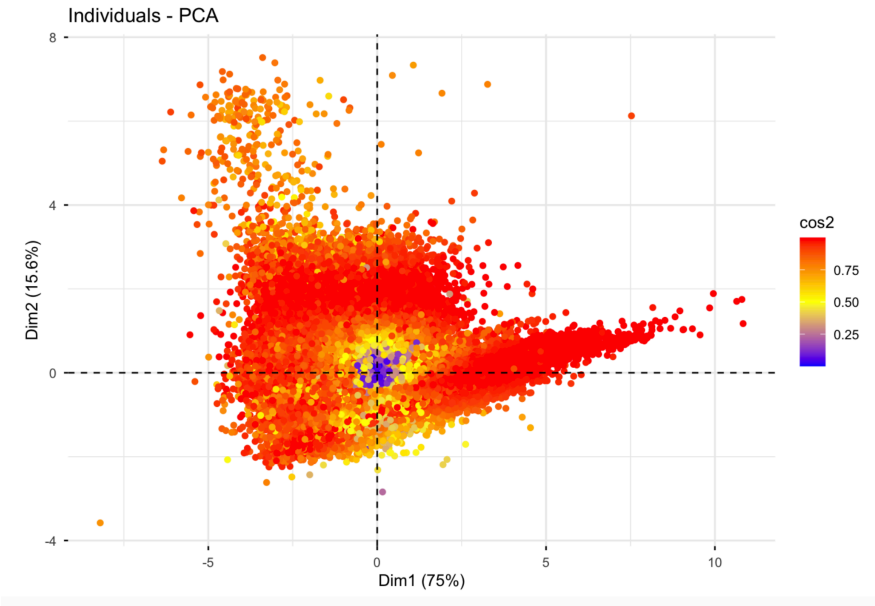

Figure 9: Quality of representation ($\cos^2$) of each feature on the first two principal components. Magnitude filters are well-represented by PC1, while redshift is captured almost entirely by PC2.